

Energy Transmission
2026-04-20

Felix Feustel

Symmetrical Components, Calculation of Short-Circuits

Contents

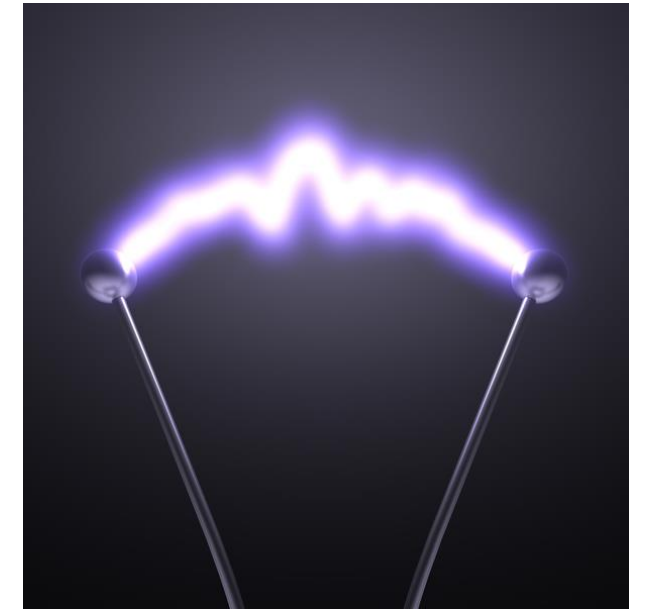
- ♦ Short-Circuits
- ♦ Symmetrical Components
- ♦ Exercises:
 - ♦ Per-unit values
 - ♦ Per-unit transformer impedance
 - ♦ Faults and symmetrical components
 - ♦ Double line-to-line fault on parallel line

Short-circuits



Short-circuit

- ◆ Accidental or intentional **conductive path** between two or more conductive parts **forcing the electric potential differences** between these conductive parts to be equal to or **close to zero**
 - Causes: Insulation failure, flashover, physical damage, human error
 - Low-ohmic connection
 - Arcing: self-maintained gas conduction
- ◆ Effects:
 - High currents
 - Induced over-voltages on neighboring equipment
 - Equipment damage (e.g due to high current)
 - Danger for humans and animals (e.g. due to arcing)



Fault analysis

- ♦ Design of protection system requires the knowledge of fault currents.
- ♦ The information obtained from the fault analysis is used for:
 - Relay settings
 - Determination of circuit breaker interrupting capacity
 - Selection of fuses and their characteristics
 - Determine fault-current distribution and busbar voltage levels during fault conditions

Line-to-line short-circuit



Example: Faults in England and Wales transmission system

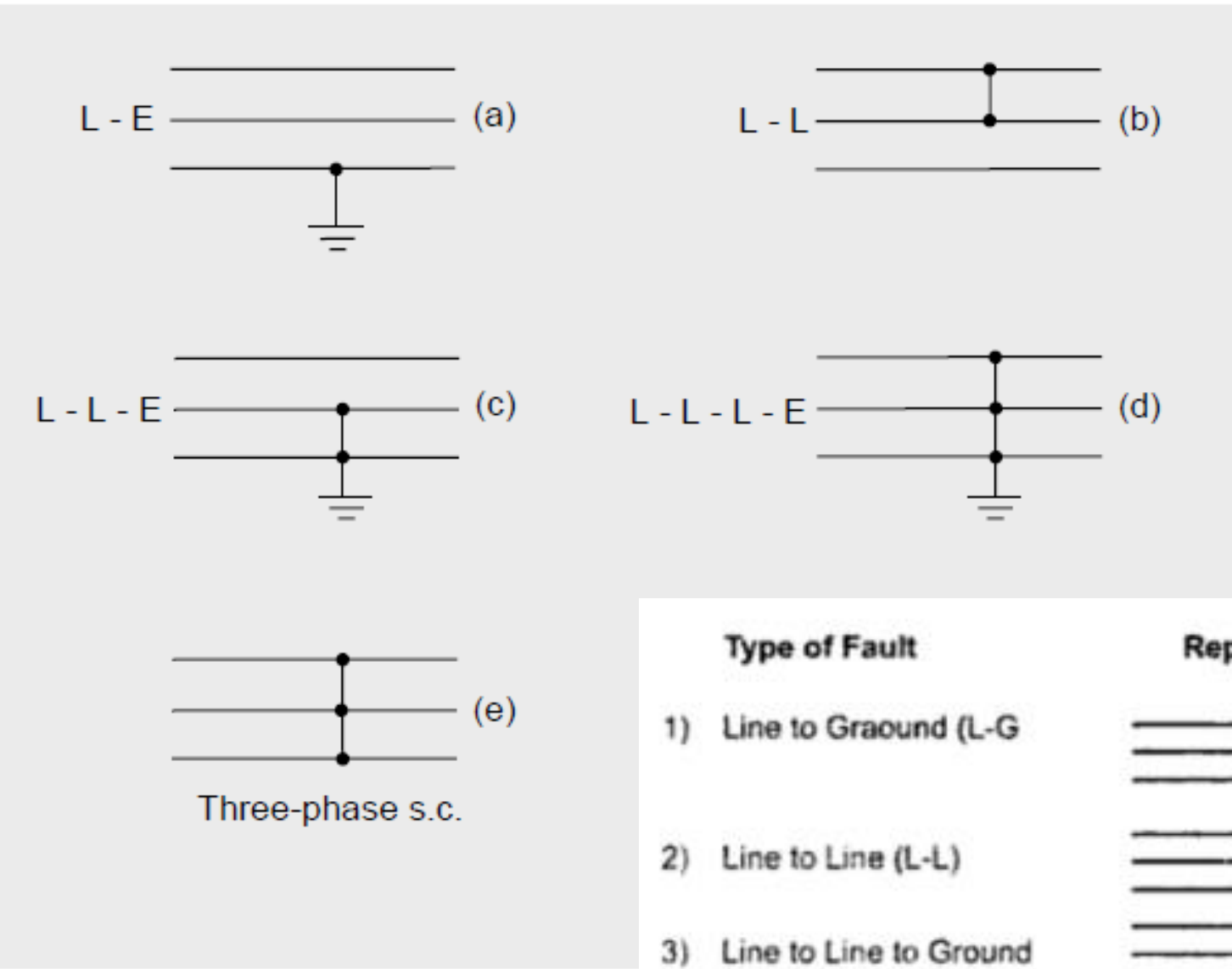
Table 7.1 Causes of overhead-line faults, England and Wales system 66 kV and above

	Faults/160 km of line/year
Lightning	1.59
Dew, fog, frost	0.15
Snow, ice	0.01
Gales	0.24
Salt spray	0.01
Total	2 faults per 160 km, giving a total of 232 faults on system/year

Table 7.2 Distribution of faults, England and Wales Transmission system

Type	Number of faults/year
Overhead lines	289
Cables	67
Switchgear	56
Transformers	59
Total	471

Types of short-circuits



Type of Fault	Representation	% occurrence
1) Line to Graound (L-G)		85
2) Line to Line (L-L)		8
3) Line to Line to Ground (L-L-G)		5
4) Line to Line to Line (L-L-L)		2 or Less

Solution methods

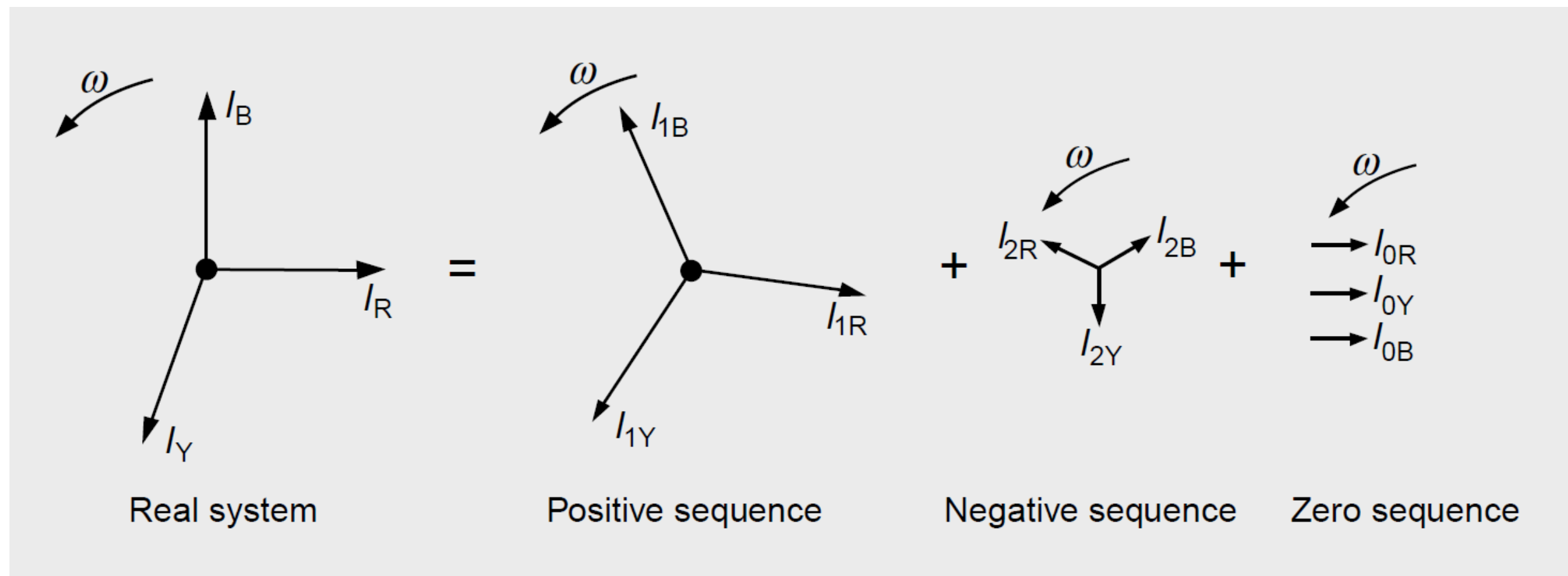
- ◆ Problem: Every non-symmetrical short-circuit (i.e. all short-circuits that are not three-phase) produces a non-symmetric three-phase system
- ◆ Solution methods (to reduce calculation complexity):
 - Nodal analysis
 - Symmetrical components

Symmetrical components



Symmetrical components

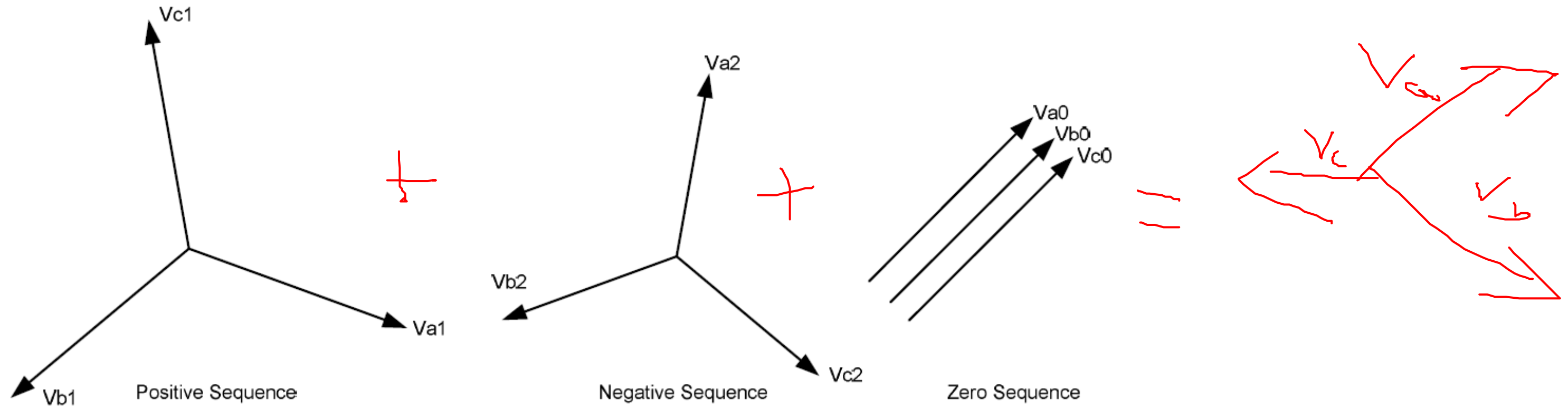
- ♦ Idea: An asymmetrical system is separated into three symmetrical parts
- ♦ Benefit: Just as on symmetrical systems, the calculation can be done on the first phase only



Symmetrical components and fault types

Type	Positive	Negative	Zero
Healthy	x		
Line-to-ground	x	x	x
Line-to-line	x	x	
Double line-to-ground	x	x	x
Three-phase	x		
Three-phase-ground	x		

Sequence voltages



◆ Express phase voltages as sequence voltages:

$$V_a = V_{a0} + V_{a1} + V_{a2} \quad (7.4a)$$

$$V_b = V_{b0} + V_{b1} + V_{b2} \quad (7.4b)$$

$$V_c = V_{c0} + V_{c1} + V_{c2} \quad (7.4c)$$

The j and a vectors

- ◆ Recall the operator j. In polar form, $j = 1\angle 90^\circ$. Multiplying by j has the effect of rotating a phasor without affecting the magnitude.

$$1 = 1.0 + j0.0$$

$$j = 1\angle 90^\circ$$

$$j^2 = 1\angle 180^\circ = -1$$

$$j^3 = 1\angle 270^\circ = -j$$

$$-j = 1\angle -90^\circ$$

- ◆ In a similar manner, the a operator is defined as unit vector at an angle of 120° , written as $a = 1\angle 120^\circ$.

$$1 = 1.0 + j0.0$$

$$a = 1\angle 120^\circ$$

$$a^2 = 1\angle 240^\circ$$

$$a^3 = 1\angle 360^\circ = 1\angle 0^\circ$$

$$1 + a^2 + a = 0$$

$$a + a^2 = -1$$

$$1 + a = 1\angle 60^\circ$$

$$1 + a^2 = 1\angle -60^\circ$$

$$a - a^2 = j\sqrt{3}$$

$$a^2 - a = -j\sqrt{3}$$

$$1 - a = \sqrt{3}\angle -30^\circ$$

$$1 - a^2 = \sqrt{3}\angle 30^\circ$$

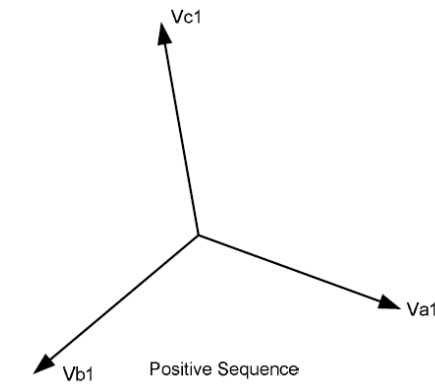
Sequence voltages (ctd.)

♦ Positive sequence:

$$V_{a1} = V_{a1} \quad (7.1a)$$

$$V_{b1} = (1\angle 240^\circ)V_{a1} = a^2V_{a1} \quad (7.1b)$$

$$V_{c1} = (1\angle 120^\circ)V_{a1} = aV_{a1} \quad (7.1c)$$

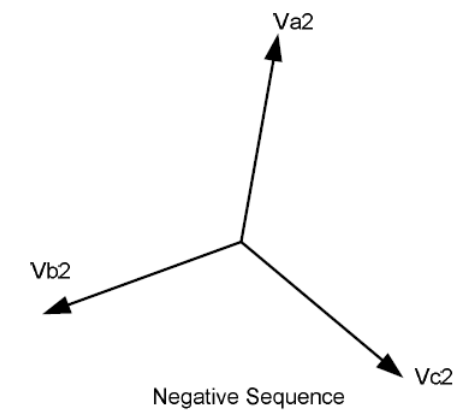


♦ Negative sequence:

$$V_{a2} = V_{a2} \quad (7.2a)$$

$$V_{b2} = (1\angle 120^\circ)V_{a2} = aV_{a2} \quad (7.2b)$$

$$V_{c2} = (1\angle 240^\circ)V_{a2} = a^2V_{a2} \quad (7.2c)$$

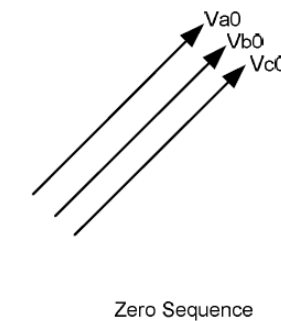


♦ Zero sequence:

$$V_{a0} = V_{a0} \quad (7.3a)$$

$$V_{b0} = V_{a0} \quad (7.3b)$$

$$V_{c0} = V_{a0} \quad (7.3c)$$



Sequence voltages (ctd.)

We may further simplify the notation as follows; define

$$V_0 = V_{a0} \quad (7.5a)$$

$$V_1 = V_{a1} \quad (7.5b)$$

$$V_2 = V_{a2} \quad (7.5c)$$

Substituting their equivalent values

$$V_a = V_0 + V_1 + V_2 \quad (7.6a)$$

$$V_b = V_0 + a^2 V_1 + a V_2 \quad (7.6b)$$

$$V_c = V_0 + a V_1 + a^2 V_2 \quad (7.6c)$$

These equations may be manipulated to solve for V_0 , V_1 , and V_2 in terms of V_a , V_b , and V_c .

$$V_0 = \frac{1}{3}(V_a + V_b + V_c) \quad (7.7a)$$

$$V_1 = \frac{1}{3}(V_a + a V_b + a^2 V_c) \quad (7.7b)$$

$$V_2 = \frac{1}{3}(V_a + a^2 V_b + a V_c) \quad (7.7c)$$

Sequence currents

It follows then that the phase current are

$$I_a = I_0 + I_1 + I_2 \quad (7.8a)$$

$$I_b = I_0 + a^2 I_1 + a I_2 \quad (7.8b)$$

$$I_c = I_0 + a I_1 + a^2 I_2 \quad (7.8c)$$

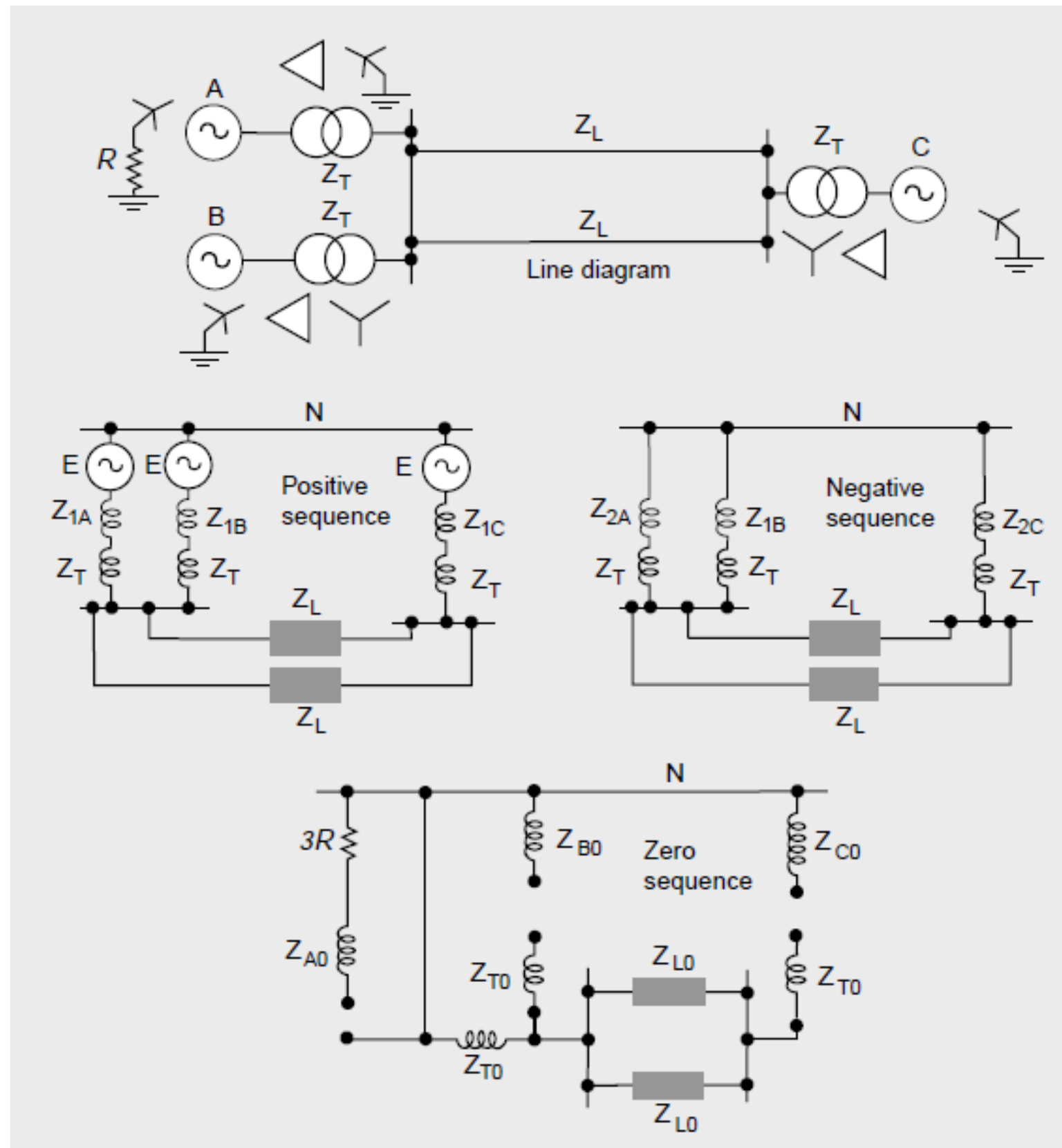
The sequence currents are given by

$$I_0 = \frac{1}{3}(I_a + I_b + I_c) \quad (7.9a)$$

$$I_1 = \frac{1}{3}(I_a + a I_b + a^2 I_c) \quad (7.9b)$$

$$I_2 = \frac{1}{3}(I_a + a^2 I_b + a I_c) \quad (7.9c)$$

Example: Transmission link and associated sequence networks



- ◆ Depending on location of the failure that is to be analyzed, networks can be reduced accordingly.
- ◆ Depending on failure type (e.g. line-to-line, line-to-ground), sequence networks will be interconnected differently

Zero sequence transformer equivalent circuits

Sequence impedances

- ◆ When an entire network is to be computed using symmetrical components, sequence impedances have to be known. If not available, formulas / equivalent circuits can be found in specialized literature.
- ◆ The impedances of electrical assets can be separated into positive, negative, and zero sequence impedances.
- ◆ For symmetrically designed assets (e.g. power lines, transformers), positive and negative sequence impedances are equal to each other.
- ◆ For rotating machines, the negative sequence reactance is smaller than the positive sequence reactance.

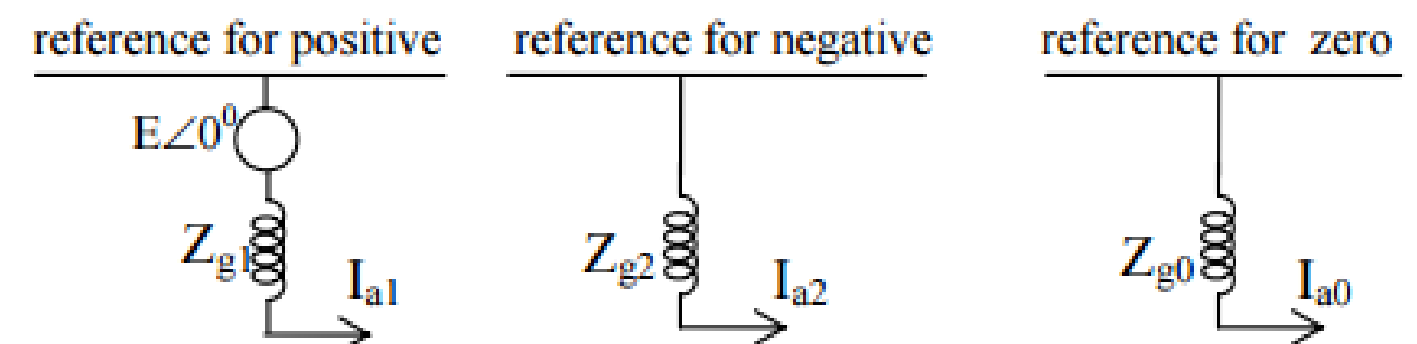
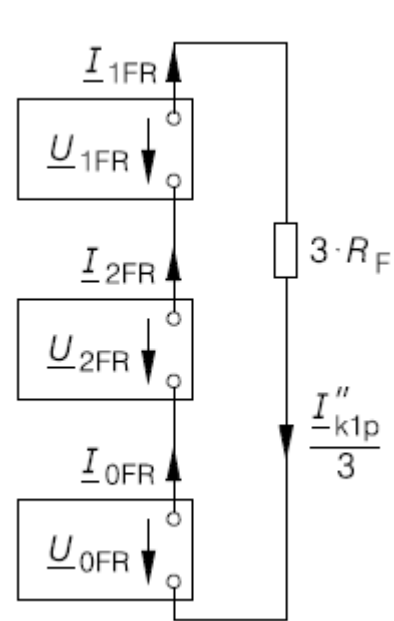
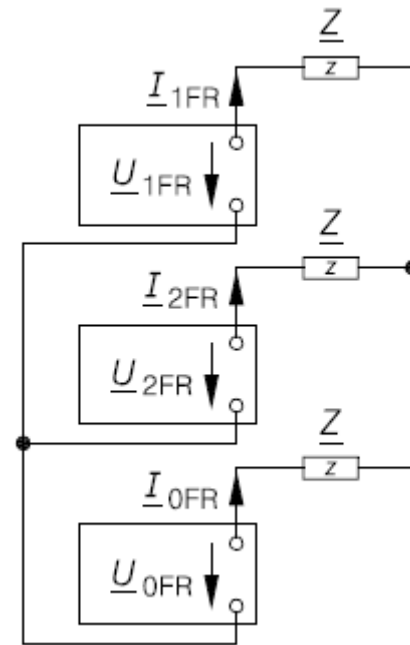


Figure 2.25 –single-line networks for sequences of generator

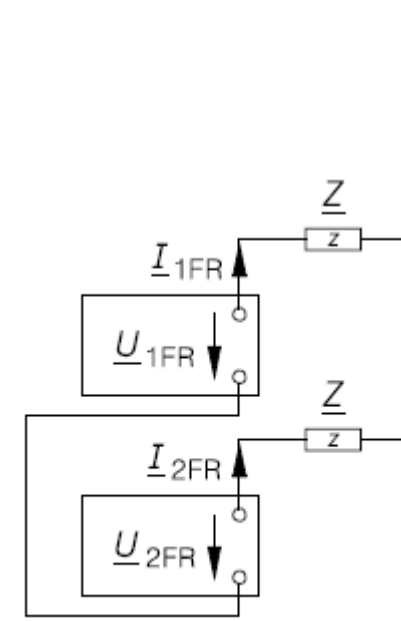
Connection of sequence components



- ◆ Phase to ground



- ◆ Two phase to ground



- ◆ Two phase

- ◆ R_F , Z : Contact resistance, impedance

Per unit

- ♦ Expression of system quantities as fractions of a defined base unit quantity to simplify calculations
- ♦ Quantities expressed as per-unit do not change when they are referred from one side of a transformer to the other

p.u. = actual value / base value

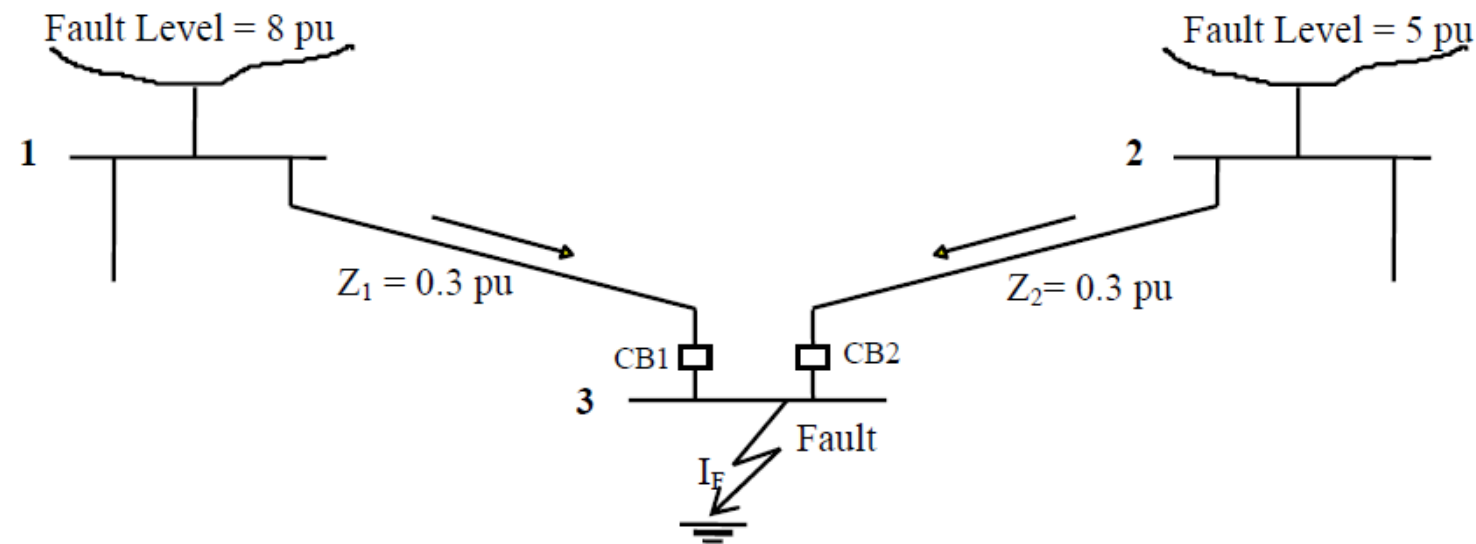
Example: $V_{base} = 138 \text{ kV}$.

Calculate per-unit voltage of $V = 136 \text{ kV}$:

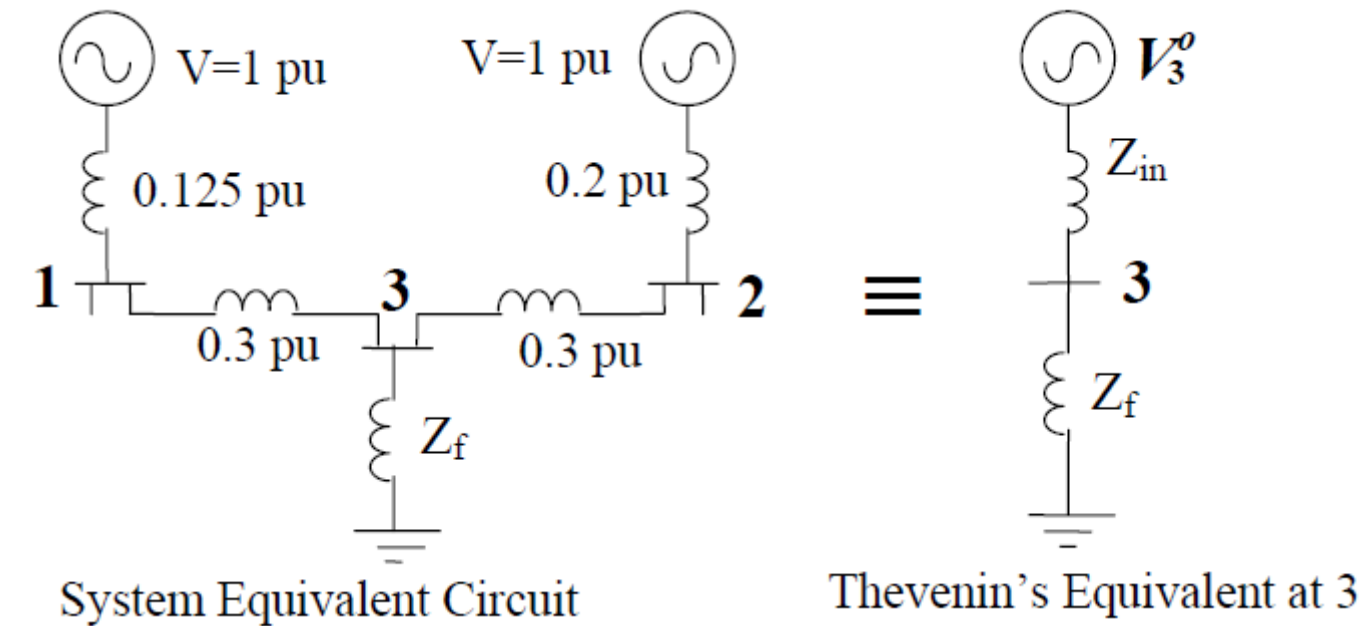
$$V_{pu} = \frac{V}{V_{base}} = \frac{136 \text{ kV}}{138 \text{ kV}} = 0.9855 \text{ pu}$$

Example: Circuit reduction for fault analysis at busbar 3

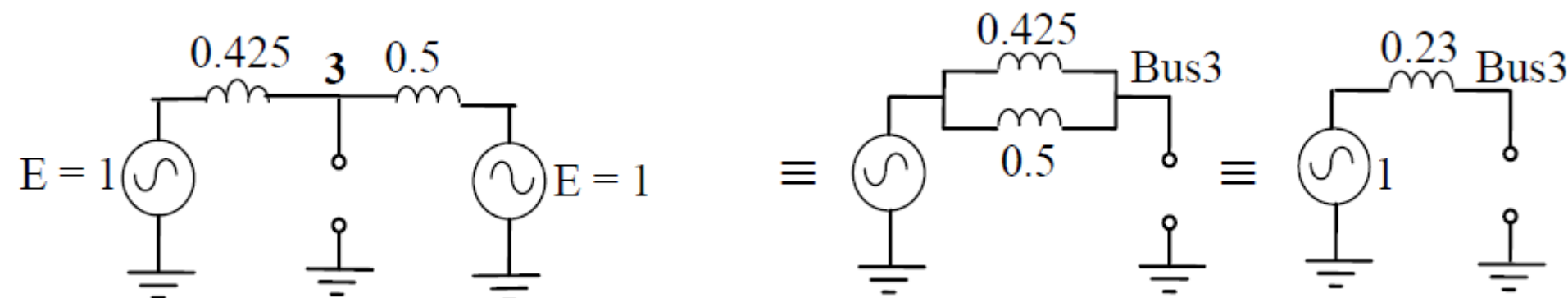
♦ Step 1: Single-line model



♦ Step 2: Equivalent circuit



♦ Step 3: Circuit for analysis (here: for busbar 3)



Exercises



Exercise 1: Per-unit values

- ♦ A system has $S_{\text{base}} = 100 \text{ MVA}$
 - a) Calculate the base current for $V_{\text{base}} = 525 \text{ kV}$
 - b) Using this value, calculate the actual line current and phase voltage where $I = 4.95 \text{ p.u.}$, and $U = 0.5 \text{ p.u.}$

Exercise 2: Per-unit of transformer impedance

- ♦ A 900 MVA, 525/241.5 autotransformer has a nameplate impedance of 10.14%
 - a) Determine the impedance in ohms, referenced to the 525 kV side
 - b) Determine the impedance in ohms, referenced to the 241.5 kV side

Exercise 3: Faults and symmetrical components

- ♦ Given $Z_0 = 0.199\angle 90^\circ$ p.u., $Z_1 = 0.175\angle 90^\circ$ p.u., $Z_2 = 0.175\angle 90^\circ$ p.u.,
 - a) compute the fault current and voltages for a single line-to-ground fault
 - b) compute the fault current and voltages for a line-to-line fault
 - c) compute the fault current and voltages for a double line-to-ground fault

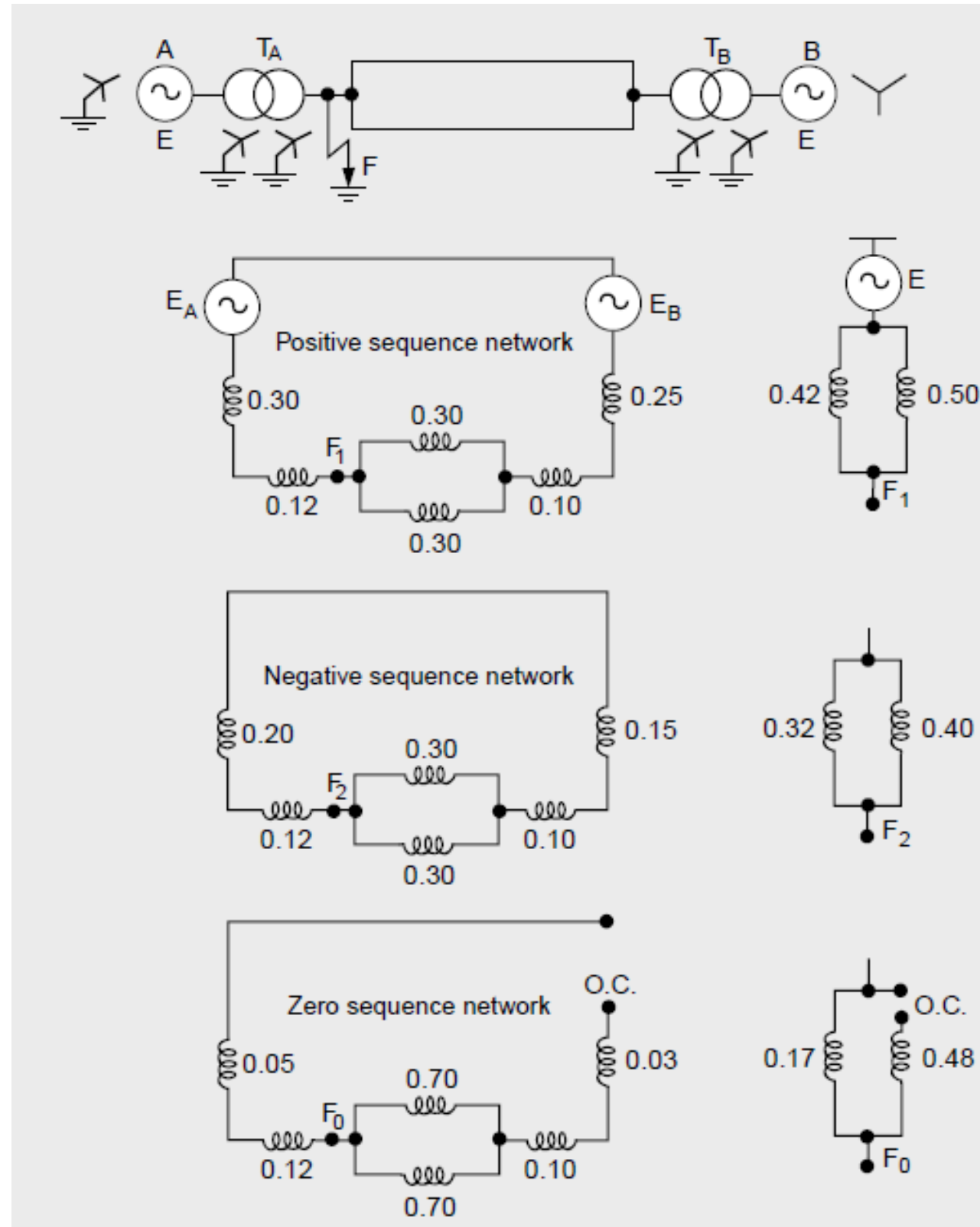
Exercise 4: Double line-to-line fault on parallel line

- ♦ A synchronous machine A generating 1 p.u. voltage is connected through a star-star transformer, reactance 0.12 p.u., to two lines in parallel. The other ends of the lines are connected through a star-star transformer of reactance 0.1 p.u. to a second machine B, also generating 1 p.u. voltage. For both transformers, $X_1 = X_2 = X_0$. The relevant per unit reactances of the plant, all referred to the same base, are as follows:
 - For each line: $X_1 = X_2 = 0.3$, $X_0 = 0.7$
 - For machine A: $X_1 = 0.3$, $X_2 = 0.2$, $X_0 = 0.05$
 - For machine B: $X_1 = 0.25$, $X_2 = 0.15$, $X_0 = 0.03$.

The star points of machine A and of the two transformers are solidly earthed.

Calculate the current fed into a double line-to-earth fault on the line-side terminals of the transformer fed from A.

Single-line diagram and sequence networks for Exercise 4



References



References

- ♦ V. Crastan, *Elektrische Energieversorgung 1* (2nd Ed., Springer, 2010), pp. 455 – 509
- ♦ K. Heuck [et al.], *Elektrische Energieversorgung*, (8th Ed., Vieweg + Teubner, 2010), pp. 570 – 599
- ♦ International Electrotechnical Commission, *IEC 60050 - International Electrotechnical Vocabulary*
- ♦ J. R. Lucas, *“Power System Faults”*, *Power System Analysis* (University of Idaho, 2005)
- ♦ S. E. Marx, *“Symmetrical Components 1 & 2”*, *28th Annual Hands on Relay School* (Washington State University 2012)
- ♦ B. M. Weedy [et al.], *Electric Power Systems* (5th Ed., Wiley, 2012), pp. 239 - 279